

Multiclass Instance Segmentation for In-Row Vineyard Navigation: A Controlled Comparison Against the Binary-Mask Baseline



UNIVERSITY OF
LINCOLN

Edosa Ebohon
30436293

30436293@students.lincoln.ac.uk

School of Engineering and Physical Sciences
College of Health and Science
University of Lincoln

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Supervisor Dr. Riccardo Polvara

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Acknowledgements

Firstly, I want to thank somebody, and somebody else. Here is another thing.

Abstract

Here is the abstract for this project report.

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Chapter 1

Introduction

1.1 Background and motivation

Autonomous robots that drive between crop rows underpin a range of precision-agriculture tasks — high-throughput phenotyping, targeted spraying, mechanical weeding — that larger over-canopy machinery cannot perform (Sivakumar et al., 2021). Each of them depends on the same primitive capability: staying in the row, reliably, for the length of a season.

That capability is limited by localisation. Satellite positioning, the default answer elsewhere in outdoor robotics, is degraded by plant canopy and by the dense vegetation of tree rows, and the error it incurs there is large relative to the clearance a robot has between the rows (Higuti et al., 2019; Navone et al., 2025). Vision is the practical alternative, and the question that follows is how the robot should represent what it sees.

This dissertation sits at the intersection of computer vision, robotics and control that defines the MSc in Robotics and Artificial Intelligence: a learned perception module, a geometric stage that converts its output into a navigable line, and a controller that converts that line into motion. The contribution is not a new architecture for any one of those stages, but a controlled measurement of how a design decision taken in the first propagates into the last.

Three terms are used throughout in their established senses (Kirillov et al., 2019). *Semantic* segmentation assigns a class label to every pixel without separating individual objects. *Instance* segmentation detects and segments each object separately. *Panoptic* segmentation combines the two. The distinction matters here

because the formulations compared in this study do not all belong to the same paradigm.

1.2 The problem

Vision-based row following has not converged on a single representation, but one recurring choice is a binary crop mask, in which every pixel is assigned either to a single foreground class or to background. de Silva et al. (2024) state the design decision plainly, representing a crop row “as a single object rather than individual plants”, and Aghi et al. (2021) carry the same representation into vineyards.

The formulation is under challenge, though not from a single direction. Recent work objects that it depends on the sky being visible above the row (Navone et al., 2025), that the free-space region it relies on degenerates under sparse vegetation (Liu et al., 2023), that pixel-wise segmentation is the wrong intermediate representation altogether (Sivakumar et al., 2024), and that learned segmentation costs more annotation and computation than it returns (A. Ahmadi, Halstead and McCool, 2022). What none of these objections addresses is the question of how many classes such a mask should carry.

That question is worth asking in a vineyard because a vineyard row is not a continuous band of foliage. It is defined by discrete rigid elements — vine trunks rooted in the ground and vertical trellis poles — which remain geometrically meaningful whether or not the sky is visible and whether or not foliage has closed over the corridor. Class identity is already exploited in vineyards on this basis, but for localisation rather than navigation, and the two published class selections disagree with one another: Papadimitriou et al. (2022) retain trunks for their stability of appearance across seasons, while de Silva et al. (2025) retain trunks and discard poles because uniform shapes frustrate feature matching. Both criteria belong to feature matching. Neither asks what a line-fitting stage needs.

The question has therefore not been tested where it would bite: downstream of perception, in a crop whose structure survives canopy closure, and under conditions where the answer can be attributed to the formulation rather than to the reference it is measured against.

1.3 Research question and scope

The study asks a single question. Does it matter, by the time a steering command is issued, whether the perception stage kept trunk and pole apart or collapsed them into one class — and if it does, by how much?

It is posed as a measurement rather than as a directional hypothesis, because either answer is useful: a practitioner deciding whether to annotate two classes or one needs to know the size of the effect, not its sign.

Three perception arms share a single downstream pipeline. The first two produce a binary mask: every pixel labelled either as row structure — trunk or pole — or as background, so that no distinction survives between the two kinds of structure. The first arm produces that mask with a semantic model, the second with an instance model. The third uses the same instance model but keeps trunk and pole as separate classes, so the distinction is preserved for the geometric stage that follows. Only the second-to-third contrast is controlled: it holds architecture, training regime and data fixed and varies the label scheme alone. The first-to-second contrast varies many things at once and is treated throughout as a comparison of complete systems rather than an attribution to architecture.

Three constraints bound the work. The evaluation is offline, against recorded robot deployments, which supports per-frame analysis but not closed-loop tracking. The reference trajectory in those recordings was produced under autonomous topological navigation (Polvara et al., 2024) rather than by a human driver, so the executed steering reflects a different sensing modality from visual row following and cannot serve as a tuning target for the visual controller; controller gains are instead derived from first principles. And the annotation scheme folds foliage into background, so foliage cannot be exploited as a navigation cue.

1.4 Contributions

This dissertation makes four contributions.

First, a controlled comparison of binary against class-aware segmentation for downstream centreline extraction in a discrete-structure crop, in which the control

is confirmed by the training behaviour of the two arms and not only by their configuration.

Second, a result that holds at two levels of measurement. Where the comparison is controlled — the same architecture and training, differing only in binary against class-aware — the arms are indistinguishable in the centreline they estimate, across four recordings spanning bare-vine and canopy conditions, and in the steering commands that follow from it. Where the comparison is not controlled, between the two binary arms, small differences do emerge on the canopy recordings, but they are smaller than the precision of the reference used to measure them. Both levels agree on which comparison shows a difference and which does not, and it is that agreement, rather than the absence of a difference alone, that carries the finding.

Third, an evaluation of availability alongside accuracy. The multiclass arm yields a usable centreline less often than either binary arm, consistently in direction across every dataset evaluated, including recordings from a later season and from a second site with different camera hardware. Measuring both outcomes gives a practitioner both halves of an answer that either alone would misrepresent: the additional class detail buys no measurable accuracy and costs availability.

Fourth, a method for comparing perception arms against a reference that is known to be inexact. The error measured against the recorded trajectory is separated into a component contributed by the reference and a component contributed by the pipeline; arms are then compared only through paired differences, in which any component common to all three cancels. Six independent groups report that no absolute reference is obtainable in this setting, and the approach is offered as a way of drawing comparisons despite that.

Chapter 2

Literature Review

2.1 In-row navigation and the localisation problem

Reliable traversal of crop rows is the capability on which a range of precision-agriculture applications rests. Sivakumar et al. (2021) identify high-throughput phenotyping, ultra-precise pesticide treatment, mechanical weeding and cover-crop planting as applications of small under-canopy robots, noting that these are not possible with the over-canopy tractors and aerial vehicles that would otherwise serve. Each depends on the robot traversing crop rows repeatedly and reliably across a growing season, so row-following is not one capability among many but a precondition for the others.

The difficulty is localisation. Real-time kinematic (RTK) satellite positioning — carrier-phase correction against a fixed base station — reaches centimetre-level accuracy in open conditions (Bakken et al., 2021), yet degrades precisely where small robots are most useful. Higuti et al. (2019) report measuring over 0.4m of error beneath a sorghum canopy using a high-end RTK-GNSS antenna, against the few-centimetre precision the same class of equipment achieves with an unobstructed view of the sky. Wang et al. (2022) add that satellite positioning degrades further under extreme weather and, more fundamentally, cannot detect obstacles in an unstructured field, so accurate positioning alone would not make navigation safe. Cost compounds the argument: Sivakumar et al. (2021) report a fiftyfold reduction in sensing cost for their monocular camera relative to the LiDAR system they compare against. Those figures are specific to their hardware

and to 2021 prices, but the cost gap they describe recurs as a motivation across this literature.

Vineyards concentrate the localisation problem. Aghi et al. (2021) note that canopy on either side of the row obstructs satellite reception, and Navone et al. (2025) identify tall, dense vegetation as the primary cause of satellite signal blockage in tree-row environments. This chapter is therefore concerned with vision-based row-following, and with a question that precedes both geometry and control: how the scene should be represented once it has been perceived.

2.2 Segmentation with a binary crop mask

Vision-based row-following has not settled on a single representation. Sivakumar et al. (2021) regress the robot’s pose relative to the row directly from the image, stating that this avoids segmenting the plants from the ground at all; their later system replaces that regression with semantic keypoints (Sivakumar et al., 2024). Liu et al. (2023) regress a continuous traversability heatmap rather than any class assignment. Winterhalter et al. (2021) build a vegetation feature map and fit line patterns to it with the Hough transform and RANSAC (Fischler and Bolles, 1981), using no learned component. A. Ahmadi, Halstead and McCool (2022) threshold an excess-green colour index, reduce each connected region to a centre point, and fit lines to those points. Segmentation is one line of work among several, not a consensus.

Within that line, the recurring representation is a binary crop mask: a per-pixel assignment of the scene into foreground and background, with everything of navigational interest collapsed into a single class. What occupies that foreground class varies. de Silva et al. (2024) train a U-Net with a binary cross-entropy objective on labels in which each row is drawn as a line of uniform width, so that “a crop row is represented as a single object rather than individual plants” — an abstraction chosen, in their words, to help the network “predict the complete crop row despite the presence of discontinuities” — before extracting the central row geometrically and steering by visual servoing. Aghi et al. (2021) carry the same representation into vineyards, segmenting each frame into a mask in which, in their words, “1 means obstacles and 0 free-space”, then summing that mask over

image columns and driving a proportional controller from the largest run of free space. Their own earlier system had reached a binary mask by a different route: Aghi, Mazzia and Chiaberge (2020) threshold stereo depth to separate near field from far, with a convolutional classifier over three steering classes as a fallback. The move to learned segmentation the following year replaced a geometric criterion for free space with an appearance-based one, so the representation was adopted rather than inherited. Nor is the formulation exclusive to learned methods: A. Ahmadi, Halstead and McCool (2022) arrive at an equivalent two-class mask through colour indexing and Otsu thresholding, without training anything. One further distinction matters for what follows. The arms compared in this study span the semantic and instance paradigms (Kirillov et al., 2019), which differ in what they assert about a scene as well as in how many classes they carry. Only one of the two contrasts available between them is controlled: the binary-to-multiclass comparison varies the label scheme alone, while the semantic-to-instance comparison varies architecture, training regime and paradigm together. The latter is treated throughout as a comparison of complete systems, not as an attribution to any single factor.

What the binary mask discards is the identity of the structures that define the row. Whether that loss matters downstream has not been tested in a vineyard, and the following section shows that the formulation is under challenge on entirely different grounds.

2.3 The formulation under challenge

The binary mask has been challenged from four directions, and the objections differ enough that they are worth separating.

The first concerns how the mask is read rather than how it is produced. Navone et al. (2025) observe that methods of this kind are “restricted to situations where the distinct separation between the plants and the sky allows for the identification of the row’s center”, and that where canopy closes over the corridor this leads to “the systematic failure of some methods”. They name the approach of Aghi et al. (2021) *SegZeros*, after its search for runs of zeros in the mask, and report its trajectory failing under vegetation coverage; in their comparison it is marked

impractical for pergola vineyards, tall tree rows and pear orchards. Their remedy, however, retains binary segmentation and changes only the search, replacing the zero-cluster criterion with a global minimum over the column histogram. The objection is to a reading of the mask, not to the mask.

The second targets the same method on different grounds. Liu et al. (2023) note that pixel clustering “relies on having only one empty space cluster, thus may not work well in several cases, e.g., early stages of growth or sparse vegetation”, and add that closing the control loop in image coordinates “limits the ability to sense distance and represent changes in elevation”. Their own system abandons class assignment altogether, regressing a continuous traversability heatmap instead.

The third is the strongest. Sivakumar et al. (2024) state that “pixel-wise segmentation of the scene as an intermediate step is not useful for the task of under-canopy navigation”, concluding that a traversable region “cannot be obtained by just segmenting the scene because of the challenges posed by occlusion and clutter”. Two qualifications bound the claim. The comparison is against excess-green colour indexing and a general-purpose foundation model applied without adaptation, rather than a supervised model trained for the task; and the setting is under-canopy corn and soybean, where the row is a continuous mass of leaves and residue rather than a sequence of structures.

The fourth is economic rather than representational. A. Ahmadi, Halstead and McCool (2022) concede that learned methods “are often more accurate than their traditional counterparts” but reject them because they “require a significant amount of labeled data and more computational resources”, preferring colour indexing. That objection applies to any learned perception module and bears on annotation cost, which is precisely what a multiclass scheme increases.

Taken together, the two objections that bite hardest — sky dependence and cluster degeneracy — are both about recovering *free space* from a mask, and both fail in the same circumstance: when the region between the rows stops being visually distinct.

2.4 Discrete-structure crops

That shared failure mode suggests where the exception lies. A pipeline that fits geometry to the detected structures defining the row, rather than clustering the free space between them, is not exposed to either objection: structures do not require sky contrast to be visible, and they do not degenerate into a single undifferentiated region when vegetation thins. The question of how many classes such a pipeline should carry is separate from, and left open by, both objections. Whether a crop offers such structures is therefore the question, and the literature answers it directly.

Higuti et al. (2019) draw the distinction while arguing the opposite case. Explaining why orchard methods do not transfer to their own crops, they write that corn and sorghum differ from orchards because “instead of well-defined tree trunks in orchards, weeds and hanging leaves can be easily mistaken with or even occlude the stalks” of those crops. Sivakumar et al. (2021) reinforce the point from the same research programme that later supplies the strongest objection above, noting that orchard row spacing is an order of magnitude wider than in commodity crops and that heuristic line fitting remains possible there where it fails under canopy. The objection to segmentation and the limit on its scope originate in the same body of work.

Vineyards sit on the structured side of that division. The site studied here is the Ktima Gerovassiliou estate near Epanomi, Greece, an area of more than 100 ha of which Polvara et al. (2024) used five adjacent corridors for a season of robot deployments. Papadimitriou et al. (2022), collecting separately at the same estate, report consecutive vine trunks roughly 0.8 m apart. The row is defined by two kinds of rigid element — vine trunks rooted in the ground, and vertical trellis poles — alongside a horizontal water pipe running its length.

Persistence is not unconditional, and both papers are candid about its limits. Mapping the estate across the season with lidar, Polvara et al. (2024) report that “poles disappear from the map because the plants are now covered by leaves”; Papadimitriou et al. (2022), working at the same estate, measure trunk segmentation quality falling from 0.723 intersection-over-union in winter to 0.625 in summer. Yet Polvara et al. (2024) also identify trunks and poles as the “stable

temporal features” that survive when everything else in the scene has changed. The structure degrades under canopy; it does not disappear.

The open question is therefore narrower than whether such structure exists. It is whether preserving its *identity* through the perception stage — as distinct classes rather than as undifferentiated foreground — changes anything by the time a steering command is produced.

2.5 Class-aware perception in vineyards

Class identity is already exploited in vineyards, though for localisation rather than for navigation, and the two published choices disagree about which classes matter.

Papadimitriou et al. (2022) restrict feature extraction to trunk regions, segmenting them with Mask R-CNN and computing visual words only inside the resulting masks, “since they slightly vary among seasonal alterations”. The criterion is appearance stability: a trunk photographed in winter should still be recognisable in summer. They select that architecture after comparing a semantic model against an instance one on their own data, finding the instance model better on every visit and by the widest margin in summer.

de Silva et al. (2025) build the six-class annotation set used in this study and then make the opposite decision about poles. Enriching keypoint descriptors with semantic context, they retain trunks and buildings, but “pipes and poles are excluded due to uniform shapes that often produce mismatches”. The criterion here is descriptor distinctiveness: a class whose instances look alike is a liability when the task is to match one instance to another across frames.

Both criteria belong to feature matching, and neither transfers to row following. A row-fitting stage does not need to tell one pole from another; it needs enough of them, in the right places, to constrain a line. Uniformity is irrelevant to that task, and availability — which neither criterion considers — is decisive. The classes have been selected twice, on grounds that do not apply to the problem considered here.

Outside vineyards, the closest published work is Bakken et al. (2021), who train

a segmentation over crops, lanes and background from labels projected automatically from a robot’s satellite-referenced trajectory. Their concern, however, is the source of the labels rather than the choice of classes: no binary alternative is trained, and the scheme is never compared against one. What class structure buys the downstream stages is not a question the study asks.

2.6 Evaluating navigation against an imperfect reference

A comparison of perception formulations at the level of navigation output requires a reference for that output, and this is where the field is weakest.

Wang et al. (2022) explain why cross-study comparison is not available. Agricultural datasets, they write, “come from different environments, aim at different objects, are small in scale, and lack the integration of multiple scenarios and tasks”, which “make it difficult to use them as benchmarks of algorithm evaluation”. Methods are therefore compared across studies that share neither data nor metric, and a controlled comparison within a single study is the only route to an attributable result.

Within studies, the reference itself is unreliable, and the papers reviewed here say so with unusual candour. Higuti et al. (2019) abandon lateral offset as their evaluation metric because “inconsistencies originate from multiple sources other than the perception itself”, measuring lane width instead. Winterhalter et al. (2021) survey markers by hand and estimate their attainable accuracy at 3° and 5 cm, adding that “since these measurements are determined relative to naturally grown plants a more precise position cannot clearly be defined”. Navone et al. (2025) report that ground truth was “unavailable due to the complexity and demanding nature of measurement” and substitute a lidar-derived lateral displacement. A. Ahmadi, Halstead and McCool (2022) drive the robot manually and use the logged satellite positions as the “correct” path, their own quotation marks acknowledging what that assumes. In Bakken et al. (2021) and Liu et al. (2023) the same instrument generates the training labels and serves as the evaluation reference.

Six independent groups, none of them with an absolute reference, and each working around its absence differently. Any comparison drawn against such a reference inherits its error.

2.7 The gap

Three conditions have to hold together for the question of segmentation formulation to be answerable, and no existing study satisfies all three.

The question must be asked *downstream*. The vineyard work that varies class structure does so for localisation, where the selection criteria — appearance stability and descriptor distinctiveness — have no bearing on whether a line can be fitted through the result. The navigation work that reaches a steering command does not vary class structure at all.

It must be asked in a crop where structure survives. The published objections to binary segmentation concern the recovery of free space, and both fail where the corridor stops being visually distinct; a discrete-structure crop is where an alternative representation could in principle answer them.

And it must be asked under conditions where the answer can be attributed. Cross-study comparison is unavailable for want of a common benchmark, and within-study comparison rests on a reference that six independent groups report as unobtainable in absolute terms.

This study addresses that conjunction: a controlled comparison of segmentation formulations, carried through to centreline geometry and steering command, in a trellised vineyard, with the validity of the comparison established rather than assumed.

2.8 Aim and objectives

The aim of this study is to measure what preserving trunk and pole identity through the perception stage is worth, in the terms a navigation system is judged by: the accuracy of the centreline it estimates, the smoothness of the steering it commands, and how often it produces an estimate at all. Where no effect is

found, the aim extends to bounding its size, so that indistinguishability is not confused with insufficient measurement. Six objectives follow.

1. **Establish a controlled three-arm perception baseline.** Train a semantic binary arm, an instance binary arm and an instance multiclass arm on a common scene-level split; verify empirically that the binary-to-multiclass contrast varies only the label scheme, and document the respects in which the semantic-to-instance contrast does not; and report each arm in its native metric across multiple training seeds. Where no shared benchmark exists (Wang et al., 2022), a controlled within-study comparison is the only route to an attributable result, and the semantic and instance paradigms are formally distinct (Kirillov et al., 2019), so their metrics are not commensurable.
2. **Quantify the effect of formulation on centreline geometry.** Implement a single geometric stage shared by all three arms; estimate paired cross-arm differences in lateral offset and heading, with interval estimates, across recordings spanning bare-vine and canopy conditions; and measure how often each arm yields a centreline estimate at all. Accuracy and availability are separable outcomes: a formulation that is no less accurate may still produce an estimate less often, and a practitioner needs both halves of that answer.
3. **Quantify the effect at the command level.** Derive a steering controller from the platform’s kinematics, generate commands from each arm’s centreline estimate, and estimate paired cross-arm differences in command smoothness under an identical control law. A perception difference is operationally meaningful only if it survives into the command issued, so the geometric result requires an independent second level of measurement.
4. **Characterise the operating envelope and its failure mode.** Determine how the pipeline behaves on frames outside the in-row condition it was designed for — during turns, transitions between rows, and while stationary — quantify the error it produces there, and measure how much of that behaviour can be suppressed by filters that reject unreliable estimates, and at what cost to valid ones. Row switching and headland turning —

manoeuvres in the turning area at the row ends — are identified as unmet needs by Wang et al. (2022) and left as future work by Higuti et al. (2019); Winterhalter et al. (2021) identify the mechanism, in that row detectors return a best-fitting pattern for any input, including one containing no row structure at all.

5. **Establish the validity of the comparison against an imperfect reference.** Separate the error measured against the reference into the part contributed by the reference itself and the part contributed by the pipeline, identify which components affect all three arms identically, and show that comparing the arms against one another removes them. Since the reference is not an exact measure of the row centre, the comparison is only meaningful if the reference’s own error cancels in that subtraction.
6. **Assess behaviour beyond the training distribution.** Evaluate the arms on recordings from a later season and year at the same site, and on recordings from a second site with different camera hardware, and determine which findings reproduce and which do not. Sivakumar et al. (2021) show growth-stage shift to be severe and asymmetric, so a result established on one recording condition is not yet a result about the formulation.

Each objective is discharged by having made a measurement, not by a measurement falling in a particular direction. Chapter 5 returns to them in turn.

Chapter 3

Methodology

3.1 Research design

This study is a controlled comparative experiment rather than a system development exercise. The distinction determines everything that follows: the object under test is a design decision taken inside a perception module, and the pipeline surrounding that module exists to make the decision’s downstream consequences measurable.

The comparison is drawn within a single study because no alternative is available. Wang et al. (2022) report that agricultural navigation datasets “come from different environments, aim at different objects, are small in scale”, which makes them unsuitable as shared benchmarks; two published accuracy figures obtained on different sites, with different references and different metrics, cannot be differentiated to isolate the effect of a label scheme. Holding the site, the recordings, the downstream pipeline and the evaluation fixed, and varying only the perception formulation, is the only arrangement in which a difference can be attributed to that formulation.

Three arms are required rather than two. Every binary approach in the reviewed literature uses a semantic architecture, which assigns a class to each pixel without separating objects; a class-aware alternative is most directly expressed by an instance architecture, which separates them. Comparing the two would change the labels and the architecture at once, and a difference between them could not be attributed to either. The third arm resolves this — an instance model trained on the same binary labels. It differs from the class-aware arm in its labels alone, and from the semantic arm in its architecture alone.

Two alternative designs were considered. A closed-loop field trial would measure what a practitioner ultimately cares about, but it is not available here: the recordings are fixed, so the same conditions cannot be presented to three arms in turn, and the steering executed during them was generated by a different sensing modality altogether. A simulated environment would supply exact ground truth and would permit closed-loop operation, but its fidelity becomes part of the result — Aghi et al. (2021) report simulated trajectory error roughly a third of the lowest figure any of the reviewed work obtains in a real vineyard, so a formulation difference could be suppressed or created by the simulator rather than observed. Simulation is treated here as a route to the closed-loop evaluation this study cannot perform, and is returned to in Chapter 7.

Each objective set out in Section 2.8 is specified in one of the sections that follow.

Table 3.1: Where each objective is specified.

Objective		Specified in
1	Controlled three-arm baseline	Section 3.3
2	Effect on centreline geometry	Sections 3.4, 3.5
3	Effect at the command level	Section 3.6
4	Operating envelope and failure mode	Section 3.5
5	Validity against an imperfect reference	Section 3.7
6	Behaviour beyond the training distribution	Section 3.2

3.2 Data

Two kinds of data are used, and the relationship between them requires care. The perception arms are trained and evaluated on an annotated image corpus, reduced to the two classes described below; the comparison itself is drawn on recorded robot deployments, which carry no annotations. Quantities that require ground truth — segmentation overlap, detection precision and recall — are therefore available only on the annotated corpus, and the arms are compared on the deployment recordings through the output of the pipeline they feed rather than through the quality of their masks.

The annotated corpus is distributed as 1,035 images, but these are not 1,035 independent observations. 805 of them are augmented variants, generated non-uniformly, so a single scene may recur several times under altered brightness,

exposure or blur. Deduplicating by source scene leaves **230 unique scenes**, and it is that figure the design rests on. Six classes are annotated — trunk, pole, pipe, building, robot and vehicle — of which two are used. Trunks and poles are the structures that define the row geometrically; pipe, building, robot and vehicle are assigned to background, so that the multiclass arm carries two foreground classes rather than six. A reader encountering “multiclass” should therefore expect two.

The corpus was partitioned at the scene level in a 70/20/10 ratio — 161 training, 46 validation and 23 test scenes — stratified by canopy state, so that bare-vine and canopy conditions are proportionally represented in each partition. Splitting at the scene level rather than the image level is necessary because of the augmentation: a frame-level partition would place a brightness- or blur-shifted copy of a training image into the test set. The ratio follows conventional practice for computer-vision datasets of this scale.

The deployment recordings come from the campaign described in Section 2.4, and span bare-vine and canopy conditions within one season, together with recordings from a later season and from a second site. Some of the annotated scenes were extracted from the same recordings used for evaluation. Frames overlapping a located annotated scene are therefore excluded from the evaluation set: without that exclusion, the downstream comparison would in part be drawn on frames the perception arms had been trained on, and any difference between them could reflect memorisation rather than generalisation.

3.3 The three arms

The semantic arm is a U-Net (Ronneberger, Fischer and Brox, 2015) with a ResNet-34 encoder initialised from ImageNet, trained with a composite objective of cross-entropy and a soft Dice term in the family introduced by Milletari, Navab and S.-A. Ahmadi (2016), applied per class and averaged. The generalised Dice variant of Sudre et al. (2017), which weights classes by inverse squared volume, was considered and rejected as an over-correction for a two-class problem. The two instance arms are YOLOv11 segmentation models (Jocher, Qiu and Chaurasia, 2025), initialised from COCO and trained under the framework’s own composite objective. No peer-reviewed description of the YOLOv11 segmenta-

tion head exists; its mask branch belongs to the family introduced by Bolya et al. (2019), in which instance masks are produced by linearly combining a set of image-wide prototype masks with per-instance coefficients.

Between the two instance arms, everything is held fixed except the labels: architecture, initialisation, optimiser, schedule, augmentation, batch size, epoch count and random seeds are identical, and the only substantive difference in configuration is the path to the label set. This is the contrast the study turns on, and it invites an immediate objection. The multiclass arm’s default downstream configuration pools trunk and pole detections before fitting, and the binary arm produces a single pooled class to begin with. If both end in a pooled set of detections, what distinguishes them?

The difference lies in the model rather than in what follows it. The binary arm is trained to answer one question about each candidate — whether it is row structure. The multiclass arm is trained to answer two — whether it is row structure, and which kind. A network optimised to separate trunks from poles must learn features that tell them apart, and those features shape every detection it emits: where the box falls, how confident the model is, whether a thin distant pole is detected at all. Pooling the classes afterwards discards the label, not the effect that learning the label had on the detections.

This is visible in the training record rather than argued from first principles. Over a hundred epochs the two arms’ localisation, mask and distribution-focal loss terms track one another to within a hundredth, while the classification term diverges by six to eleven times and remains higher for the multiclass arm at every epoch. The arms learn the same thing about where structure is and different things about what it is. The controlled variable is therefore isolated in fact, not only by construction.

The contrast between the semantic and the instance binary arms is not controlled in the same way. The two differ in thirteen respects: architecture, pre-training corpus, segmentation paradigm, optimiser, learning rate — by a factor of one hundred — weight decay, schedule, epoch count, batch size, loss function, early-stopping patience, augmentation policy, and output representation. Each arm was trained under settings conventional for its own framework, and no attempt

was made to equalise them. The consequence is stated once and held throughout: that contrast compares two complete perception systems, and a difference between them cannot be attributed to any single factor, architecture included.

3.4 The shared pipeline

Every arm’s output passes through the same eight-stage geometric pipeline. Only the extraction of ground-contact points from detections is arm-specific; every stage after it is identical. Perception produces detections; each detection contributes a *base point*, the bottom-centre of its bounding box, taken as the point at which the structure meets the ground. The instance arms yield one base point per detected instance; the semantic arm, which produces no instances, yields one per connected component of foreground above a minimum area. Base points are assigned to the left or right side of the corridor by their horizontal position in the image, projected onto the ground plane, and fitted per side.

A side is fitted only if it carries at least two base points within a near-field window ahead of the robot. This is a count criterion, not a distance one: detections beyond the window contribute to the fit but cannot seed it. The requirement follows from the geometry of the problem: a line through a single point is undetermined. It accounts for the large majority of the frames on which the pipeline declines to answer.

Fitting therefore ends in one of four states. Where both sides fit and a centreline can be derived from them, the frame yields an estimate. Where only one side fits, or where neither does, the pipeline emits nothing: no offset and no heading. A fourth state, in which both sides fit but the centreline derivation fails, did not occur at all on the primary recording and is reported only for completeness.

Emitting nothing in the second and third states is a design decision rather than a limitation, and an alternative exists: where one side has been fitted, a centreline could be placed at half the known corridor width from it. That variant was specified but deliberately not implemented: it substitutes an assumption about corridor geometry for a measurement, and any error in the assumption would enter the estimate without being visible in the output. pipeline declines to answer is therefore measured and reported rather than treated as failure.

The centreline is evaluated at a fixed look-ahead distance ahead of the robot rather than at the robot itself, following the convention of pure-pursuit steering, in which a controller tracks a point some distance along the intended path. Evaluating there rather than averaging over the visible range also removes a systematic bias that the projection incurs under a small sensor tilt.

Two safeguards protect the fit from individual bad detections, which matter because a line through a handful of points is sensitive to any one of them and the line is the pipeline’s output. The first addresses a failure mode specific to instance segmentation, whose mask head can emit a single very large mask covering foliage rather than a structure (Bolya et al., 2019): detections occupying more than a set fraction of the frame are discarded before projection. The threshold was set against observation — the largest legitimate detection in the bare-vine recording, a trellis pole close to the camera, occupies roughly a tenth of the frame, so a limit of fifteen per cent removes the pathological case while leaving genuine structure untouched. The second operates within the fit: each side’s points are compared against the median of that side, and those outside a tolerance of it are excluded before the line is fitted. A median is used rather than a mean because it is insensitive to the few extreme values being guarded against.

3.5 What is measured

Four quantities are recorded for every arm on every recording, and they answer different questions.

Coverage is the proportion of eligible frames on which the pipeline emits a centreline at all. Eligible means recorded while travelling within a row, and not overlapping an annotated scene the arms were trained on; frames recorded while turning at the row ends or while stationary are excluded, since the pipeline was never intended to serve them. A method that is accurate on the frames it answers but answers few of them is not equivalent to one that answers most.

Lateral error is the signed distance between the estimated centreline and the reference at the look-ahead, in metres, and is the primary measure: lateral displacement determines whether the robot passes cleanly between the vines or strikes

one, whereas a heading error matters only through the displacement it accumulates. *Heading error* is reported alongside it throughout.

Command smoothness is the root-mean-square change in commanded turn rate between consecutive frames. Where the pipeline emits nothing, the controller holds its previous command rather than stopping, and a held command changes by zero — so frames on which nothing was estimated appear perfectly smooth. Two forms are therefore reported. The inclusive form covers every in-row frame and describes what the robot would have experienced; the exclusive form covers only frames carrying a fresh estimate and describes how smooth the perception-driven commands are in themselves. An arm that declines often is flattered by the first, and the gap between the two measures how much.

3.5.1 Frames the pipeline was not built for

The measurements above are confined to frames recorded while travelling within a row, which is the condition the pipeline was designed for. A robot working a field does not spend all its time there: it stops, it reaches the end of a row and turns, and it crosses between rows. Winterhalter et al. (2021) observe that a row detector will return its best-fitting pattern for any input given to it, including one containing no row structure at all, so a pipeline that is never asked what it does outside its design condition has not been fully characterised.

Frames outside the in-row condition are therefore separated into three categories — stationary, turning at a row end, and in transition between rows — and the pipeline is run on them unchanged. Two things are recorded: how often it emits an estimate where it should emit none, and how far that estimate lies from the path the robot actually drove, which is the only reference available when there is no row to be centred in.

Two filters are then evaluated for their ability to suppress those outputs. The first uses the robot’s own motion, which distinguishes travelling along a row from turning or standing still without reference to the image. The second uses the geometry of the estimate itself, rejecting centrelines whose offset, heading or implied corridor width fall outside the range observed in-row. Each is assessed on two counts: how much of the unwanted output it removes, and how many valid

in-row estimates it discards in doing so. A filter that rejected everything would score perfectly on the first and is worthless on the second.

3.5.2 The class-priority sweep

One further comparison is made within the multiclass arm alone. The three-arm design asks whether preserving class identity changes the navigation output; it cannot ask what each class is individually worth to a line fit, because the other two arms carry no class distinction to vary. That second question is the one Section 2.5 leaves open, where two published selections retain and discard poles on grounds drawn from feature matching and neither considers what a row fit requires.

Every detection carries a class, and each side of the corridor needs base points to fit a line through; the sweep varies which detections are used. Fifteen configurations are evaluated over the same stored detections, so no model is retrained and what is compared is the selection rule alone.

One configuration is class-agnostic: every detection is used, trunk and pole together, as the binary arms do in effect. Two are hard filters, using one class and discarding the other entirely. The remaining twelve prefer one class but fall back on the other where there is too little of it — on each side of the corridor, if that side carries at least a threshold number of the preferred class, only that class is used; if it carries fewer, both are used together. Six thresholds are tried for each of the two classes, spanning the range of per-side detection counts observed in the recordings.

One configuration is carried forward to the main comparison, chosen on validation data by lowest lateral error among those meeting a minimum coverage requirement. A tie-break was fixed before the sweep was run: if the best configuration's interval overlapped that of the class-agnostic one, the class-agnostic configuration would be kept, on the ground that a difference which cannot be distinguished does not justify the added complexity. That situation arose on both bare-vine recordings: the lowest-error qualifying configuration was a threshold variant, its interval overlapped class-agnostic, and class-agnostic was therefore retained. It should be said plainly that this is not the same as class-agnostic performing best.

On the canopy recordings the question does not arise at all, since no configuration — class-agnostic included — meets the coverage requirement there. Fixing the rule beforehand, and reporting all fifteen results rather than the chosen one, prevents the threshold from becoming a parameter adjusted until the outcome improves.

3.6 The control strand

A difference in perception matters operationally only if it survives into the command the robot acts on. The geometric measurements answer whether the arms estimate the same centreline; they do not answer whether a controller fed those estimates would behave differently. The controller is therefore included as an instrument of measurement rather than as a contribution: it is identical for all three arms, so any difference in what it emits is attributable to what it was given. This separation of learned perception from an analytical controller follows the modular arrangement of Sivakumar et al. (2021), who adopt it because an analytical low-level controller generalises readily across varying terrain.

The obvious way to set such a controller’s gains is to fit them against the steering the robot actually executed during the recordings. That was attempted and abandoned. The recordings were made under autonomous topological navigation with satellite positioning (Polvara et al., 2024), so the robot was not steering by looking at the row. Fitting the controller to that steering would tune it to imitate a different sensing system.

Two measurements confirm it. The perceived centreline explains well under one per cent of the variation in the executed turn rate, so the two are almost unrelated. And a search across a grid of candidate gains, validated on held-out passes, settled on the smallest gains available — a controller that barely steers at all, which is the best way to match a signal carrying no usable information.

Gains are instead derived analytically from the platform’s kinematics under a small-angle approximation, with the two free parameters — how quickly the robot should converge on the row centre, and how much overshoot to permit — fixed against the behaviour observed in the recordings rather than fitted to them. The derivation is given in Chapter 4.

A third possibility, tuning by driving the system to sustained oscillation and reading the resulting period, is the classical procedure of Ziegler and Nichols (1942) and remains standard practice (Åström and Hägglund, 2006). It is unavailable here for the same reason a closed-loop trial is: the procedure requires driving the robot and observing how it responds, and the recordings are fixed. This is a limitation of offline evaluation, not a judgement on the method.

3.7 Statistical treatment

Three decisions govern how the arms are compared.

Arms are compared by differencing, not by their absolute errors. On each video frame where two arms both produced an estimate, the difference between those two estimates is taken, and it is those per-frame differences that are summarised. The reference is not an exact measure of the row centre, and whatever error it carries enters every arm’s absolute figure alike — so under subtraction, any error the arms share cancels. Comparing absolutes would measure the reference as much as the arms.

Results are reported as confidence intervals, not significance tests. A confidence interval states a range within which the true difference plausibly lies; a significance test states only whether a threshold was cleared. Wasserstein and Lazar (2016) argue the latter is a poor substitute for the former, and the argument applies with force here, where the differences of interest are small. What matters is not whether a difference clears a threshold but whether the interval is narrow enough to rule out a difference large enough to matter.

The intervals are computed by bootstrapping in blocks. Bootstrapping constructs an interval by resampling the observed data many times and recording how much the result varies across those resamples. Drawn one at a time, the frames would be treated as though each carried fresh evidence. They do not: at the recorded travelling speed of roughly 0.7 m s^{-1} and fifteen frames per second, a dozen consecutive frames span little more than half a metre and show almost the same scene, so an error in one recurs in its neighbours. Counting them as independent would make a few hundred genuinely distinct views look like several thousand, and the resulting interval would be correspondingly too narrow —

differences would appear resolved that the data cannot resolve. Resampling in contiguous blocks avoids this by keeping neighbouring frames together (Künsch, 1989). The block spans the distance over which one view stops resembling the next, measured from the recordings themselves by how quickly image features cease to match between frames, and is set at twice that distance so each block comfortably covers the dependence. Where a recording yields too few paired frames to contain enough independent blocks, no interval is reported for it.

Each arm is trained three times under different random seeds, and results are reported as a mean with the spread across those runs. That spread sets the scale against which cross-arm differences are read: a difference between two arms smaller than the difference between two training runs of the same arm is not a property of the arm.

Two further conventions apply. Differences are reported as proportions of the satellite positioning accuracy published for these recordings (Polvara et al., 2024), which is the tightest the reference could possibly be. That accuracy differs between sessions, from under a centimetre to several; the figure for the primary recording is used throughout so that proportions remain comparable, and the per-session figures are reported so that any comparison can be rescaled. A difference below that bound is beyond what this measurement can resolve. And where a comparison cannot support an interval at all, what is reported instead is whether the difference points the same way on every recording — a weaker statement, and one limited by the fact that the same trained models are evaluated throughout.

Chapter 4

Implementation

This section of the dissertation may vary significantly in both structure and content, depending on the type of project you are undertaking.

The precise structure should be discussed with your supervisor, but some suggestions for additional sections are given below. The key thing to note here is that irrespective of the project type, you should *justify* the choices you've made, rather than simply choosing based on expediency or familiarity. The main types of projects a student will do on the CMP9140 Research project module are below:

- Software Development Projects
- Research-focused projects not involving human participants
- Research Projects involving human participants

Its possible for a single project to adopt aspects of one or more of the types of project listed above.

4.1 Software Development Projects

If the primary deliverable of your project is a software product, then you should consider subsections detailing your approaches to the following:

- Toolsets and machine environments (i.e., the software and hardware used)
- Design (e.g., UML diagrams, database schema, prototypes)
- Testing (i.e., the types of testing used)

This list is not exhaustive. For example, a games design project may include a game design document. However, it must be noted that if your project contains

significant software development work, then most if not all of these sections should be present.

4.2 Research projects not involving human participants

For some projects, the main deliverables may come in the form of experimental results. For example, a project comparing several different algorithms may require little in the way of code, but require considerable experimentation and data analysis. As such, all methodological choices made should be documented here. Examples include:

- Dataset acquisition and annotation
- Algorithm/model design and selection
- Parameter tuning
- Performance metrics

Again, this list is not exhaustive, and you should still include relevant sections pertaining to the software artefact listed in Section 4.1 if applicable.

4.3 Research projects involving human participants

For projects involving human participants, you will need to consider a hypothesis or research question that your project will answer. You will also need ethical approval using the LEAS system. In addition to the sections outlined in Section 4.1 if you are also developing software, you may also need to provide details of:

- Participant recruitment
- Evidence that ethical procedures have been followed
- Study design (including hypotheses/research question as appropriate)
- Quantitative/Qualitative analysis (i.e., how you'll analyse the raw data)

If your study involves data collection by means of questionnaires, you may also wish to specify the questions here (or refer to them in an appendix)

Chapter 5

Results & Discussion

This section should present the findings of your work, and discuss them in the context of your original aims objectives. For software-oriented projects, how well does it meet your original requirements? Provide data where possible, e.g., results of user testing, performance measures, etc. For research-oriented projects, you should present the data in an appropriate format (tables, charts, visualisations) and provide a critical discussion around the results that provides insight and direction for future work.

Chapter 6

Legal, Social, Ethical and Professional Issues

Placeholder text for the LSEPI chapter.

Chapter 7

Conclusion

This is another relatively short chapter that is an opportunity to reflect on the project as a whole. Discuss the limitations and successes of the project, highlighting opportunities for future work.

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